

Q1

|                                                                                                                                                         |          |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--|
| 1. (a) The glass cup absorbs thermal energy (from the hot water) /<br>Heat is lost to the glass cup.                                                    | 1A       |  |
|                                                                                                                                                         | 1        |  |
| (b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$<br>$H = 168 \text{ (J } ^\circ\text{C}^{-1}) \approx 170 \text{ (J } ^\circ\text{C}^{-1})$ | 1M<br>1A |  |
|                                                                                                                                                         | 2        |  |
| (c) (i) Infra-red (radiation) / thermal radiation                                                                                                       | 1A       |  |
|                                                                                                                                                         | 1        |  |
| (ii) Remote control / infra-red (IR) thermometer /<br>night vision camera                                                                               | 1A       |  |
|                                                                                                                                                         | 1        |  |

Q2

|                                                                                                                                                         |          |                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------|
| 2. (a) $s = ut + \frac{1}{2}at^2$<br>$0 = 5t + \frac{1}{2}(-9.81)t^2$<br>$t = 1.0194 \text{ s} \approx 1.02 \text{ s}$ or $0 \text{ s}$ (rejected)      | 1M<br>1A | Or<br>$v = u + at,$<br>$0 = 5 + (-9.81)(t/2)$<br>$t = 1.02 \text{ s}$ |
|                                                                                                                                                         | 2        | Accept: $t = 1.00 \sim 1.02 \text{ s}$                                |
| (b) speed of the bead at its maximum height = $2 \text{ m s}^{-1}$<br>$\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$<br>$= 0.02 \text{ J}$          | 1M<br>1A |                                                                       |
|                                                                                                                                                         | 2        |                                                                       |
| (c) The bead and the trolley have the same horizontal velocity<br>and thus travel the same horizontal distance (within the same time<br>interval $t$ ). | 1A<br>1A |                                                                       |
|                                                                                                                                                         | 2        |                                                                       |

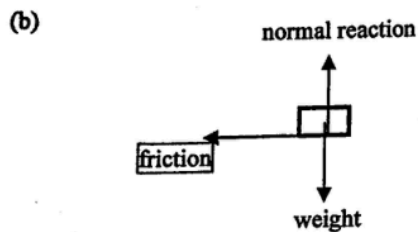
Q3

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at  $t = 4$  s.
- (b) Displacement = total area of the graph  
 $= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10 - 4)$   
 $= 25$  m  
 Position of the object is 25 m from point  $O$  on the right side.
- (c) Net force acting on the object  
 $= ma$   
 $= 0.4 \times \frac{15 - (-10)}{10 - 0}$   
 $= 1$  N (towards right)

|    |    |                                                                                                         |
|----|----|---------------------------------------------------------------------------------------------------------|
| 1A | 2  |                                                                                                         |
| 1M | 1A | Or<br>$v^2 = u^2 + 2as$<br>$15^2 = (-10)^2 + 2 \times \frac{15 - (-10)}{10 - 0} \times s$<br>$s = 25$ m |
| 1A | 2  |                                                                                                         |
| 1M | 1A |                                                                                                         |
| 1A | 2  |                                                                                                         |

Q4

4. (a)  $\omega = \frac{2\pi}{60}$   
 $= \frac{\pi}{30}$  or  $0.105$  (rad s<sup>-1</sup>)

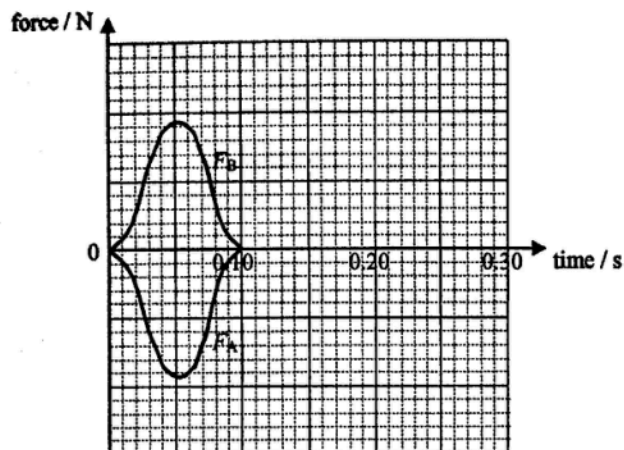


- (c) The required centripetal force  $F_C = m\omega^2 r$ , where  $r$  is the distance of plasticine  $Q$  from the clock's centre. As  $m$  and  $\omega$  are constant, smaller  $r$  gives smaller centripetal force  $F_C$ .  
 The claim is correct.

|       |   |                        |
|-------|---|------------------------|
| 1A    | 1 |                        |
| 1A    | 1 | Accept: adhesive force |
| 1A+1A | 2 |                        |

towards the right taken as positive),  
give trolley *A* a push and let it collide head-on with trolley *B*.

Force-time graphs:



The force acting on (the less massive) trolley *B*, i.e.  $F_B$ , is always equal (in magnitude and opposite in direction) to that acting on trolley *A*, i.e.  $F_A$ , as they are an action and reaction pair / according to Newton's 3<sup>rd</sup> law.

1A

2A

1A

4

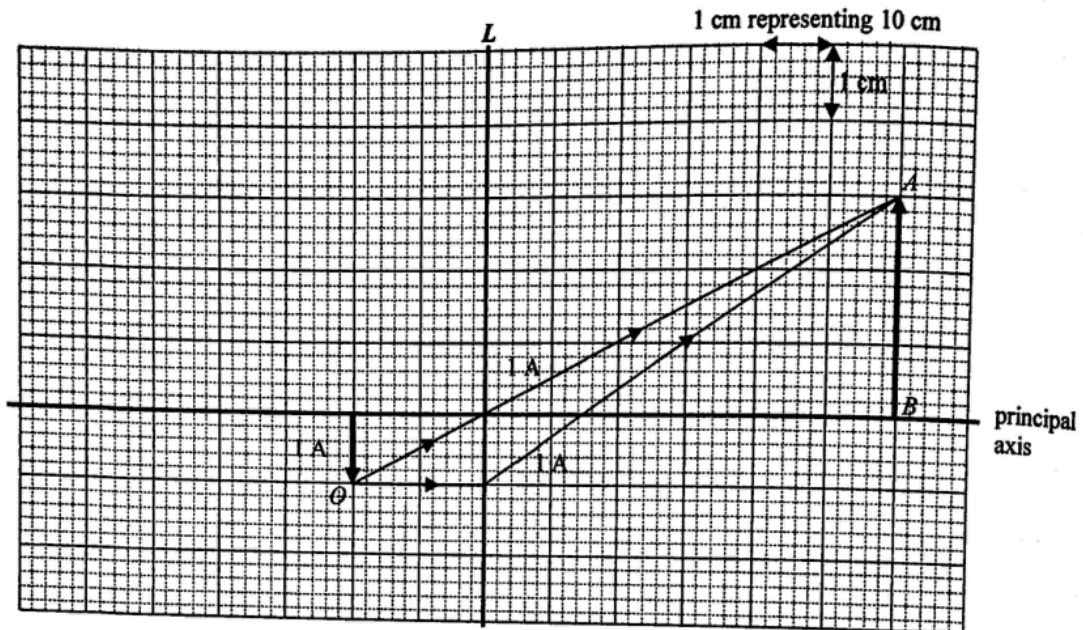
6. (a) Convex lens  
because only a convex lens can form a real image.

1A  
1A

Or  
the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.

2

- (b) (i)(ii)



- (iii) Ray diagram method:  
 $f = 15 \text{ cm}$

1A

Accept: 14 cm ~ 16 cm

Or

$$\text{By } \frac{v}{u} = 3,$$

$$\frac{0.6}{u} = 3, \text{ i.e. } u = 0.2 \text{ m}$$

$$\text{By } \frac{1}{f} = \frac{1}{u} + \frac{1}{v},$$

$$\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$$

$$f = 0.15 \text{ m or } 15 \text{ cm}$$

- (c) Move / adjust the support / smartphone closer to lens L.

- (d) The box should be painted black inside.

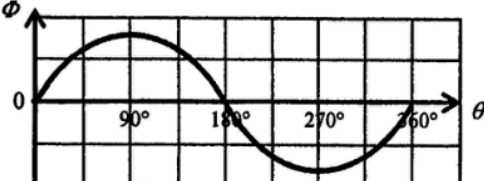
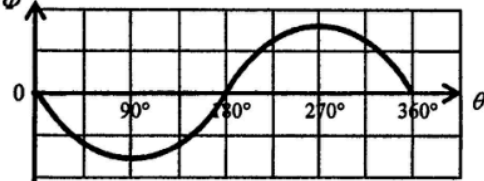
1A

1A

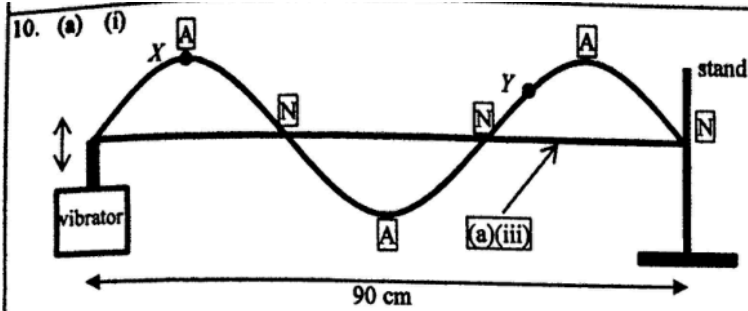
Q7

|                                                                                                                                                    |    |                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------------------------------------------|
| 7. (a) To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat.                                     | 1A |                                                  |
|                                                                                                                                                    | 1  |                                                  |
| (b) $R = \frac{V}{I} = \frac{3.5}{0.7}$<br>$= 5.0 \Omega (> 4.7 \Omega)$                                                                           | 1M |                                                  |
| The extra resistance ( $0.3 \Omega$ ) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply. | 1A |                                                  |
|                                                                                                                                                    | 1A |                                                  |
|                                                                                                                                                    | 3  |                                                  |
| (c) $I = 0.25 \text{ A}$ at $l = 30 \text{ cm}$                                                                                                    |    | Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$ |
| $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$                                                                                                       | 1M | e.c.f. from (b)                                  |
| $= 9.0 \Omega$                                                                                                                                     | 1A | Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$  |
|                                                                                                                                                    | 2  |                                                  |

Q8

|                                                                                     |    |                           |
|-------------------------------------------------------------------------------------|----|---------------------------|
| 8. (a)                                                                              |    |                           |
|    | 2A |                           |
| OR                                                                                  |    |                           |
|  |    |                           |
|                                                                                     | 2  |                           |
| (b) (i)                                                                             |    |                           |
| $\phi = BA \cos 0^\circ = 2.70 \times 10^{-4}$                                      |    |                           |
| $B \times (9.00 \times 10^{-4}) \times 1 = 2.70 \times 10^{-4}$                     | 1M |                           |
| $B = 0.300 \text{ (T)}$                                                             | 1A |                           |
|                                                                                     | 2  |                           |
| (ii)                                                                                |    |                           |
| $\varepsilon = \left  -N \frac{\Delta \phi}{\Delta t} \right $                      |    |                           |
| $= \left  -100 \times \frac{2.70 \times 10^{-4}}{0.01} \right $                     | 1M |                           |
| $= 2.70 \text{ V}$                                                                  | 1A |                           |
|                                                                                     | 2  |                           |
|                                                                                     |    | Accept: $-2.70 \text{ V}$ |
|                                                                                     | 2  |                           |

|                                                                                                                                                                                                                                                                                                                        |               |                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(i) <math>1.01 \times 10^5 \times V = 1 \times 8.31 \times 273</math><br/> <math>V = 2.2462 \times 10^{-2} \text{ m}^3 \approx 2.25 \times 10^{-2} \text{ m}^3</math></p>                                                                                                                                           | 1M<br>1A<br>2 | Accept: $0.0220 \sim 0.0225 \text{ m}^3$                                                                                                                                           |
| <p>(ii) Number of molecules in <math>1 \text{ m}^3</math><br/> <math display="block">= \frac{N_A}{\text{volume of one mole of molecule}}</math> <math display="block">= \frac{6.02 \times 10^{23}}{2.246 \times 10^{-2}}</math> <math display="block">= 2.680120 \times 10^{25} \approx 2.68 \times 10^{25}</math></p> | 1M<br>1A<br>2 | e.c.f. from (a)(i)<br>Or<br>$pV = nRT$<br>$1.01 \times 10^5 \times 1 = n \times 8.31 \times 273$<br>$n = 44.5$<br>$N = 44.5 \times 6.02 \times 10^{23}$<br>$= 2.68 \times 10^{25}$ |
| <p>(b) <math>pV = \left( \frac{N}{N_A} \right) RT</math><br/> <math>2.75 \times 10^{-10} \times 1 = \left( \frac{1.00 \times 10^{11}}{6.02 \times 10^{23}} \right) \times 8.31 \times T</math><br/> <math>T = 199.2178 \text{ (K)} \approx 199 \text{ (K)}</math></p>                                                  | 1M<br>1A<br>2 | Accept: $199 \sim 200 \text{ K}$                                                                                                                                                   |



- (ii) Particle  $X$  vibrates with a larger amplitude than particle  $Y$  while they are vibrating in phase.

- (iii) Correct sketch of a straight string

- (iv) Original wavelength

$$\lambda_1 = \frac{0.90}{1.5} = 0.60 \text{ m at } f_1 = 100 \text{ Hz}$$

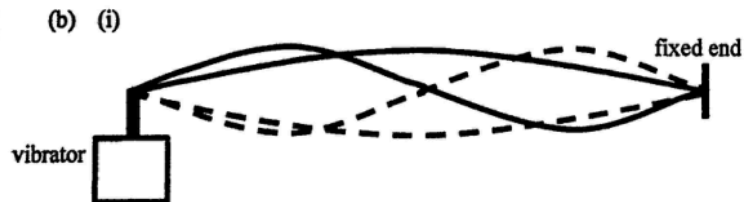
For the next stationary wave pattern,

$$\lambda_2 = \frac{0.90}{2} = 0.45 \text{ m}$$

By  $v = f\lambda$ ,

$$100 \times 0.60 = f_2 \times 0.45$$

$$f_2 = 133.33 \text{ Hz} \approx 130 \text{ Hz}$$



- (ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms.

- (iii) Any TWO:  
stationary vs travelling  
transverse vs longitudinal  
energy confined in a string vs energy being transferred through air

2A

2

1A

1A

2

1A

1

1M

1A

2

1A

1

1A

1A

2

2A

2

Or

$$f_2 = 100 \times \frac{4}{3}$$

$$= 133.33 \text{ Hz} \approx 130 \text{ Hz}$$

$$= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$$

$$(ii) \quad E_{out} = (680 \times 10^6) \times (24 \times 3600) \\ = 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$$

$$(iii) \quad \text{Efficiency} = \frac{E_{out}}{E_{in}} \times 100\% \\ = \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\% \\ = 40.8\% \approx 41\%$$

$$(b) \quad (i) \quad P = VI \\ 680 \text{ MW} = 400 \text{ kV} \times I \\ I = 1700 \text{ A} = 1.7 \text{ kA}$$

$$(ii) \quad \text{Power loss} = I^2 R \\ = (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3) \\ = 13410 \text{ W} \approx 13.4 \text{ (kW)}$$

(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.

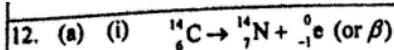
|    |   |
|----|---|
| 1M | 2 |
| 1A |   |
| 2  |   |
| 1M |   |
| 1A |   |
| 2  |   |
| 1M |   |
| 1A |   |
| 2  |   |
| 1A |   |
| 1A |   |
| 2  |   |

e.c.f. from (a)(i)(ii)

Accept: 41% ~ 42%

e.c.f. from (b)(i)



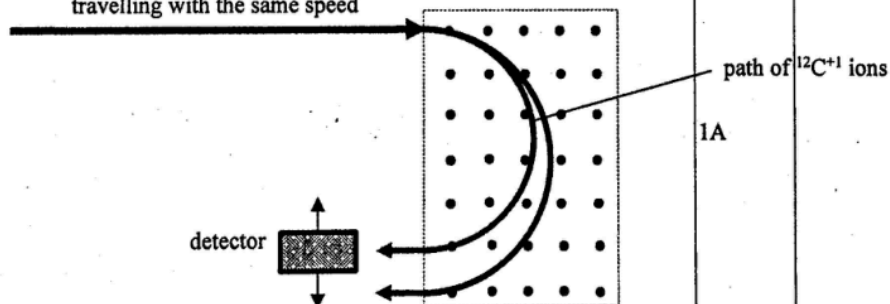


(ii)  $k = \frac{\ln 2}{t_{\frac{1}{2}}}$   
 $= \frac{\ln 2}{5370}$   
 $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}\text{)} \approx 1.21 \times 10^{-4} \text{ (yr}^{-1}\text{)}$

(iii)  $C = C_0 e^{-kt}$   
 $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$   
 $t = 6764.7 \approx 6800 \text{ (years)}$

(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.

(ii)  ${}^{12}\text{C}^{+1}$  and  ${}^{14}\text{C}^{+1}$  ions  
travelling with the same speed



(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.

1A

1

1M

1A

2

1M

1A

e.c.f. from (a)(ii)

Accept: 6760 ~ 6820 (years)

Or

$$(13.6) \left( \frac{1}{2} \right)^n = 6.0$$

$$n \log \left( \frac{1}{2} \right) = \log \left( \frac{6.0}{13.6} \right)$$

$$n = 1.181$$

$$t = n \times 5730 = 6764.7 \text{ (years)}$$

$$\approx 6800 \text{ (years)}$$

2

1A

Or

same number of protons but  
different number of neutrons

1

1A

1

1A

1A

2